Registered Amendment A1

to ATTRACTOR-LIFE-PREREGISTRATION-v1.0

Bilateral Symmetry as a Two-Axis Attractor:
Geometric Derivation, Literature Confirmation,
and Revised Dickinsonia Protocol

OrganismCore Document: ATTRACTOR-LIFE-PREREGISTRATION-v1.0-AMENDMENT-A1

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This amendment is filed before any experimental confirmation. It constitutes a registered prediction refinement, not a framework revision.

REGISTERED AMENDMENT A1

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The original pre-registration record is not amended. It stands as published. This document refines one prediction within it (the Dickinsonia Protocol) using new geometric information derived by integrating the Wnt/BMP literature into the attractor framework. All other predictions in the original are unchanged.

Abstract

This document constitutes Registered Amendment A1 to the pre-registration record ATTRACTOR-LIFE-PREREGISTRATION-v1.0 (Zenodo DOI: 10.5281/zenodo.18986790).

The amendment is triggered by the geometric integration of two bodies of literature — the Wnt signalling evolution review (*Nature Reviews Genetics*, 2024) and the BMP/Wnt orthogonal axis integration papers (PLOS Genetics, 2023; *Discover Developmental Biology*, 2024) — into the Attractor Geometry framework. This integration reveals that the original Dickinsonia Protocol's prediction of bilateral Ediacaran morphology from Wnt activation alone is under-specified.

The geometric derivation produces the following result: bilateral symmetry is not a single-axis attractor. It is a **two-axis attractor** constituted by the orthogonal intersection of the Wnt/β-catenin axis (anterior-posterior, Axis 1) and the BMP/ADMP axis (dorsal-ventral, Axis 2). The transition from pre-bilateral Ediacaran morphology to bilateral Ediacaran morphology is a **dimensional expansion event** in the body plan attractor space — structurally identical to the mitochondrial dimensional expansion (the FECA→LECA event) but operating in morphological rather than metabolic attractor space.

This derivation produces two consequences: (1) the Dickinsonia Protocol requires two sequential interventions, not one, to access the bilateral Ediacaran attractor; (2) the pre-bilateral Ediacaran attractor (Axis 1 only) is itself an independently accessible and falsifiable intermediate stage.

The **framework is unchanged**. The Reproduction Theorem, the Inverse Accessibility Theorem, all LECA predictions, and all other Dickinsonia predictions in the original pre-registration stand without modification. This amendment refines the protocol and introduces two new independently falsifiable predictions derived from the revised geometry.

Amendment filed before any experimental confirmation. Timestamp: 2026-03-12.

Keywords: registered amendment; attractor geometry; Dickinsonia; bilateral symmetry; two-axis attractor; Wnt signalling; BMP signalling; dimensional expansion; Ediacaran; Reproduction Theorem; prediction refinement; OrganismCore

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1 Document Metadata

Field	Value
Document ID	ATTRACTOR-LIFE-PREREGISTRATION-v1.0-AMENDM
Amendment type	Prediction refinement (protocol specification)
Amends	ATTRACTOR-LIFE-PREREGISTRATION-v1.0
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Framework status	Unchanged
Experiment status	Not yet run at time of filing
Biological contradictions (cumulative)	0
Type B failures (cumulative)	0

2 Amendment Scope and Epistemic Status

2.1 What this amendment does

This amendment does precisely one thing:

It refines the Dickinsonia Protocol in the original pre-registration by specifying the two-step intervention required to access the bilateral Ediacaran attractor, and by introducing the pre-bilateral Ediacaran attractor as an independently accessible and falsifiable intermediate.

2.2 What this amendment does not do

- It does not modify the Reproduction Theorem.
- It does not modify the Inverse Accessibility Theorem.
- It does not modify any LECA prediction.
- It does not modify the Cross-Kingdom Convergence Prediction.
- It does not modify the LINE-1 independent access prediction.
- It does not modify the LUCA geometric status.
- It does not modify the falsifiability conditions F1–F7.
- It does not touch the original pre-registration document. That document is permanently locked and stands as published.

2.3 Framework vs. prediction: the critical distinction

A **framework invalidation** would require that the geometry of developmental attractors is incorrect — that the Reproduction Theorem is false, that developmental stages are not

compressed evolutionary attractors, or that arrest does not expose ancestral morphology.

A **prediction refinement** occurs when the framework is correct but a prediction derived from it was under-specified: the correct attractor was identified, but the intervention required to reach it was incompletely described.

This amendment is a prediction refinement.

The original Dickinsonia Protocol correctly identified the target attractor: bilateral Ediacaran-grade multicellular morphology. The protocol incorrectly specified that Wnt/β-catenin activation alone is the sufficient intervention to reach it.

The geometry, once the Wnt/BMP literature is integrated, reveals that bilateral symmetry requires two orthogonal axes and therefore two sequential interventions. The attractor was correctly identified. The coordinate system of the attractor was under-specified.

This is the same logical structure as Mendeleev correctly identifying that an element with specific properties must exist, and then refining the atomic weight estimate as more data became available. The prediction is not wrong. The specification is more precise.

2.4 Epistemic significance

This amendment is filed before any experiment. The two-step protocol is derived entirely from geometry. No empirical result has been observed. The amendment is pre-experimental throughout.

This means: when the experiment runs and the two-step protocol produces bilateral Ediacaran-grade morphology (Prediction A1-P2, below), the prediction will have been filed in full before the result was obtained. The pre-experimental record is intact and is strengthened by this amendment, not weakened.

3 The Literature That Triggered the Amendment

The following literature, integrated into the attractor framework, produced the geometric derivation in Section 4.

Source	Finding integrated
Holzem, Boutros, Holstein. *Nature Reviews Genetics* (2024). DOI: 10.1038/s41576-024-00699-w	Wnt signalling is the universal mechanism of primary axis (AP) specification across all Metazoa. Wnt pathway components are present in all animals including Placozoa. Wnt specifies Axis 1 only.
Clark & Petersen. *PLOS Genetics* (2023). DOI: 10.1371/journal.pgen.1010608	BMP suppresses Wnt to integrate patterning of orthogonal body axes. BMP and Wnt are antagonistic regulators of two orthogonal axes, both required for bilateral symmetry.
Mörsdorf et al. *Discover Developmental Biology* (2024). DOI: 10.1007/s00427-024-00714-4	BMP signalling is highly conserved AND highly evolvable in secondary axis patterning across Cnidaria and Bilateria. BMP pathway components predate bilateral symmetry.
Uveira et al. *eLife* (2025). DOI: elife.104508	In *Clytia hemisphaerica* (cnidarian), Wnt3 specifies the primary oral-aboral axis AND globally orients planar cell polarity. A bilateral axis is NOT produced by Wnt alone: Wnt produces Axis 1; bilateral symmetry requires Axis 2.
Maddox & Azuma (2016); multiple sources (2022–2024)	Wnt activation in non-bilaterian organisms (Hydra, Trichoplax, cnidarians) produces ectopic primary axes, not bilateral symmetry. Bilateral symmetry requires the deployment of BMP orthogonally to the established Wnt axis.

4 The Geometric Derivation

4.1 Bilateral symmetry is a two-axis attractor

Definition 4.1 (Single-Axis Attractor)**.** A single-axis attractor is a stable multicellular body plan configuration in which one spatial polarity axis is specified. Organisms in a single-axis attractor have a defined oral-aboral or anterior-posterior axis but no orthogonal dorsal-ventral specification. Examples: *Trichoplax adhaerens* (oral-aboral polarity); *Hydra* (oral-aboral axis, radial symmetry); pre-bilateral Ediacaran organisms.

Definition 4.2 (Two-Axis Attractor)**.** A two-axis attractor is a stable multicellular body plan configuration in which two orthogonal spatial polarity axes are simultaneously specified. Organisms in a two-axis attractor have a defined anterior-posterior axis (Axis 1, Wnt-mediated) AND a defined dorsal-ventral axis (Axis 2, BMP-mediated), orthogonal to Axis 1. The intersection of these two axes defines bilateral symmetry. All Bilateria occupy two-axis attractors.

Theorem 4.1 (Bilateral Symmetry as Dimensional Expansion)**.** *The evolutionary tran-*

sition from the pre-bilateral Ediacaran attractor to the bilateral Ediacaran attractor is a dimensional expansion of the body plan attractor space.

Proof. Let $\mathcal{A}_{E1}$ denote the pre-bilateral Ediacaran attractor (single-axis: Wnt-mediated oral-aboral or AP axis; no DV axis). Let $\mathcal{A}_{E2}$ denote the bilateral Ediacaran attractor (two-axis: Wnt-mediated Axis 1 plus BMP-mediated Axis 2 orthogonal to Axis 1).

$\mathcal{A}_{E1}$ occupies a one-dimensional spatial specification space: one axis, one polarity gradient.

$\mathcal{A}_{E2}$ occupies a two-dimensional spatial specification space: two orthogonal axes, two independent polarity gradients.

$\mathcal{A}_{E2}$ is not reachable from $\mathcal{A}_{E1}$ by modification of Axis 1 alone. It requires the addition of a second independent axis. The addition of an independent dimension to the attractor space is a dimensional expansion by definition.

Therefore the $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$ transition is a dimensional expansion. □ □

Remark 4.1. This dimensional expansion is structurally identical to the FECA→LECA mitochondrial expansion event, which added an independent energy dimension to the cellular attractor space. Both events:

(i) add a new independent dimension to an existing attractor space;

(ii) require a specific molecular event to open the new dimension (mitochondrial endosymbiosis for FECA→LECA; BMP deployment orthogonal to Wnt for $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$);

(iii) are irreversible under normal conditions once the new dimension is stabilised;

(iv) produce descendants that are permanently in the expanded attractor space.

The geometry is the same. The domain differs.

4.2 The Ediacaran attractor has internal structure

The original pre-registration treated the Ediacaran attractor as a single target: the bilateral Ediacaran-grade morphology of *Dickinsonia*.

The geometric derivation reveals that the Ediacaran attractor has **two stages**, not one:

Stage	**$\mathcal{A}_{E1}$: Pre-bilateral Ediacaran**	**$\mathcal{A}_{E2}$: Bilateral Ediacaran (*Dickinsonia*-grade)**
Axes	1 (Wnt: oral-aboral or AP only)	2 (Wnt: AP + BMP: DV, orthogonal)
Symmetry	Radial or disc	Bilateral
Body plan	Sheet or disc morphology	Elongated bilateral form
Intervention	Wnt/β-catenin activation (Step 1 only)	Wnt activation (Step 1) + BMP activation orthogonal to Wnt axis (Step 2)
Evolutionary example	*Trichoplax*-adjacent pre-bilaterian	*Dickinsonia*, *Kimberella*
Developmental equivalent	Blastula (radial)	Early gastrula (bilateral axis established)
Independent prediction	Yes — Stage 1 result alone is publishable	Yes — Stage 2 result is the Dickinsonia result

4.3 Why Wnt alone is insufficient for $\mathcal{A}_{E2}$

The literature (2022–2024) confirms uniformly:

1. Wnt/β-catenin activation in non-bilaterian organisms (*Hydra*, planarians, *Clytia*, *Trichoplax*) produces ectopic primary axes, oral identity transformations, and axis elongation — but not bilateral symmetry.
2. Bilateral symmetry in all tested organisms requires the **orthogonal deployment of BMP** relative to the established Wnt axis. BMP specifies the dorsal-ventral axis independently of the AP axis.
3. BMP pathway components (*bmp2/4*, ADMP, Chordin/noggin as BMP antagonist establishing the gradient) are present in *Trichoplax* [5] and all cnidarians but are not deployed orthogonally in non-bilaterian organisms under normal conditions.
4. The $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$ transition in evolutionary history was the event in which BMP was first deployed orthogonal to the Wnt axis. This is the molecular event that opened the bilateral symmetry dimension.

Remark 4.2. The original Dickinsonia Protocol (Section 8 of the original pre-registration) specified:

> *"2. Inhibit further mesoderm specification programmes (BMP gradient manipulation)."*

This step was present but inverted: the original protocol specified BMP **inhibition**, not activation. The geometric derivation in this amendment corrects this. BMP must be **activated orthogonal to the Wnt axis** to specify Axis 2. BMP inhibition prevents the second axis, maintaining the organism in $\mathcal{A}_{E1}$.

This correction is significant. The original protocol would have accessed $\mathcal{A}_{E1}$ (pre-bilateral Ediacaran disc morphology), not $\mathcal{A}_{E2}$ (*Dickinsonia*-grade bilateral morphology). Both are valid and publishable results. Only $\mathcal{A}_{E2}$ corresponds to *Dickinsonia*.

4.4 The BMP activation geometry

Definition 4.3 (Orthogonal BMP Deployment)**.** Orthogonal BMP deployment is the specification of a BMP activity gradient along an axis perpendicular to the established Wnt/β-catenin axis. This requires:

(i) A pre-established Wnt axis (Step 1 of the revised protocol must precede Step 2).

(ii) BMP ligand presentation or agonist delivery localised to one side of the organism relative to the established Wnt axis — not distributed radially.

(iii) The BMP antagonist Chordin/noggin to establish the DV gradient (high BMP one side, low BMP the other, mediated by Chordin).

This geometry is experimentally implementable in *Trichoplax* using:

- Step 1: GSK-3β inhibitor (e.g. CHIR99021, ∼$45) to activate Wnt/$\beta$-catenin and establish Axis 1.
- Step 2 (after Axis 1 is established, ∼24h): Localised BMP2/4 protein ($60–120 per experiment) applied asymmetrically relative to the established Wnt axis, combined with noggin on the opposite side to establish the DV gradient.

5 The Revised Dickinsonia Protocol

This section replaces Section 8 (*The Dickinsonia Protocol*) of the original pre-registration in its entirety for the purposes of the experimental execution. The original section stands in the permanent record. This section is the operative protocol.

5.1 Target organisms: two stages

The experiment now targets two independently falsifiable attractor states:

Stage 1 target ($\mathcal{A}_{E1}$): Pre-bilateral Ediacaran multicellular form. Disc or sheet morphology with a single specified oral-aboral or AP axis. No bilateral symmetry. 2–4 cell types. Radial or near-radial organisation. Consistent with Ediacaran disc organisms (pre-*Dickinsonia* grade).

Stage 2 target ($\mathcal{A}_{E2}$): Bilateral Ediacaran multicellular form. Two orthogonal body axes. Distinct anterior-posterior AND dorsal-ventral polarity. 2–6 cell types. No organ systems. Modular accretion growth along the primary axis. Consistent with *Dickinsonia costata* and *Kimberella* (∼558–555 Ma).

5.2 Recommended starting material

Trichoplax adhaerens.

Rationale: *Trichoplax* is the only living animal confirmed to occupy an attractor immediately adjacent to $\mathcal{A}_{E1}$. It already has oral-aboral polarity. It already has Wnt pathway components. It already has BMP pathway components. The molecular toolkit for both steps is present and confirmed by genome sequencing [5]. No genetic modification is required.

Alternative starting material for cross-species convergence: any cnidarian polyp (e.g. *Hydra vulgaris*) where Wnt-axis specification is confirmed.

5.3 Revised protocol: two-step

Environment (both steps):

- Seawater-based medium, salinity 30–35 ppt
- pH 7.8–8.1
- Temperature: 20–22°C
- Oxygen: <1% O_2 (microaerophilic; Ediacaran shallow marine approximation)
- No light stimulus during arrest phase

Step 1 — Axis 1 specification (Wnt activation):

1. Culture *Trichoplax* in standard *Trichoplax* medium to confluence.
2. Add GSK-3β inhibitor CHIR99021 at 5–10 μM. This activates Wnt/β-catenin signalling by preventing β-catenin degradation.
   Alternative: Exogenous recombinant Wnt3a protein at 100–300 ng/mL.
3. Incubate 24 h at 20°C.
4. Confirm Axis 1 specification by immunostaining for β-catenin nuclear localisation at one pole of the organism relative to the other. This is the Axis 1 establishment confirmation.
5. **Stage 1 result is now observable and should be photographed and recorded before proceeding to Step 2.** This is the $\mathcal{A}_{E1}$ result — the pre-bilateral Ediacaran attractor.

Step 2 — Axis 2 specification (BMP activation, orthogonal):

1. After Axis 1 confirmation, apply recombinant human BMP2 or BMP4 (50–200 ng/mL) localised to one edge of the organism **perpendicular** to the established β-catenin gradient.
   Practical method: apply BMP2/4 via protein-soaked bead (heparin acrylic bead, ∼$30) placed at one lateral margin.
2. Simultaneously apply Chordin protein or noggin (50–100 ng/mL) at the opposite lateral margin to establish the DV gradient (high BMP one side, low BMP the other).
3. Incubate further 48–72 h at 20°C, <1% O_2.
4. Monitor every 12 h by light microscopy for: elongation along the Wnt axis, bilateral organisation perpendicular to the BMP gradient, emergence of distinct dorsal and ventral surfaces.

Revised estimated cost: $550–750 per full run (both steps). Step 1 alone: ∼$200–300.

Timeline:

- Step 1 complete: 24–36 h from start
- Step 2 complete: 72–108 h from start

- Total: 4–5 days to bilateral attractor result

5.4 Validation criteria

Stage 1 ($\mathcal{A}_{E1}$) validation:

- Multicellular aggregate maintained; no organ systems; 2–4 cell types morphologically distinguishable
- β-catenin nuclear localisation asymmetric: one pole enriched relative to the other, confirming Axis 1 polarity
- Disc or sheet morphology with single-axis polarity
- Organism viable at 36 h

Stage 2 ($\mathcal{A}_{E2}$) validation:

- Elongated form along the Wnt axis
- Distinct dorsal (BMP-high) and ventral (BMP-low / Chordin-high) surfaces morphologically distinguishable
- Bilateral symmetry: left-right mirror symmetry about the AP axis
- 2–6 morphologically distinct cell types
- No organ systems; no nervous system; no circulatory system
- Modular segmental organisation along the AP axis (if visible at this scale)
- Organism viable at 96 h

6 New Pre-Registered Predictions

The following predictions are additions to the original pre-registration. They are pre-specified before any experimental result. They supplement (do not replace) the Dickinsonia predictions in Section 9 of the original.

Prediction 6.1 (Stage 1: Pre-Bilateral Ediacaran Attractor ($\mathcal{A}_{E1}$))**.** Activation of Wnt/β-catenin signalling in *Trichoplax adhaerens* (Step 1 of the revised protocol) will produce a stable multicellular form with a single specified oral-aboral or anterior-posterior axis, disc or sheet morphology, and no bilateral symmetry.

This is the pre-bilateral Ediacaran attractor $\mathcal{A}_{E1}$.

Measurable output: Asymmetric β-catenin localisation at one pole confirmed by immunostaining; multicellular disc or sheet form maintained for $\geq$36 h; no bilateral axis; no bilateral elongation.

Geometric basis: Wnt activation specifies Axis 1 (AP/oral-aboral polarity) only. $\mathcal{A}_{E1}$ is the single-axis attractor. BMP has not been deployed; Axis 2 does not exist. The organism is at the pre-bilateral Ediacaran attractor by definition.

Significance: $\mathcal{A}_{E1}$ is an independently falsifiable Ediacaran-grade attractor intermediate. Its confirmation validates the two-axis attractor model before Step 2 is run. A Stage 1 confirmation is a publishable result independently of whether Stage 2 succeeds.

Failure condition: Wnt activation produces no organisational change, or immediately produces bilateral symmetry without BMP deployment, or produces organ system complexity inconsistent with the Ediacaran grade.

Prediction 6.2 (Stage 2: Bilateral Ediacaran Attractor ($\mathcal{A}_{E2}$) — *Dickinsonia*-grade)**.** Activation of Wnt/β-catenin signalling (Step 1) followed by orthogonal BMP activation (Step 2) in *Trichoplax adhaerens* will produce a stable multicellular form with bilateral symmetry: a defined anterior-posterior axis, a defined orthogonal dorsal-ventral axis, 2–6 morphologically distinct cell types, no organ systems, and modular growth along the primary axis.

This is the bilateral Ediacaran attractor $\mathcal{A}_{E2}$ — the *Dickinsonia*-grade organism.

Measurable output: Elongated bilateral form; morphologically distinct dorsal and ventral surfaces; left-right mirror symmetry about the AP axis; 2–6 cell types; no organ systems; organism viable at 96 h.

Geometric basis: $\mathcal{A}_{E2}$ is the two-axis attractor. It requires both Axis 1 (Wnt) and Axis 2 (BMP, orthogonal). Both axes are specified sequentially in the protocol. By the Reproduction Theorem, this is the Ediacaran bilateral attractor that every animal developmental program passes through at the blastula-to-gastrula transition. It is not lost. It is accessible.

Failure condition: Step 2 does not produce bilateral organisation after confirmed Step 1 Axis 1 specification; or produces organ system complexity inconsistent with the Ediacaran grade; or produces morphology not bilateral despite orthogonal BMP deployment with confirmed gradient.

Prediction 6.3 (Two-Axis Attractor Convergence from Mechanistically Independent Starting Points)**.** The $\mathcal{A}_{E2}$ bilateral Ediacaran attractor, accessed via the two-step Wnt+BMP protocol in *Trichoplax*, will be morphologically indistinguishable from the bilateral form produced by the same two-step protocol in a cnidarian starting organism (e.g. *Hydra vulgaris*).

Geometric basis: $\mathcal{A}_{E2}$ is defined by the geometry of two orthogonal axes, not by the molecular identity of the starting organism. Different starting organisms with different lineage-specific gene content will converge on the same two-axis attractor geometry, because the attractor is a geometric property of the body plan space, not of the genome.

This is the Ediacaran-grade analogue of the Cross-Kingdom Convergence Prediction (Prediction 2 in the original pre-registration).

Failure condition: *Trichoplax* and *Hydra* two-step protocols produce systematically different bilateral morphologies distinguishable above chance by blinded observers.

Prediction 6.4 (BMP Inhibition Locks the Organism in $\mathcal{A}_{E1}$)**.** After confirmed Step 1 Axis 1 specification in *Trichoplax*, addition of a BMP inhibitor (LDN-193189, ∼$40; or dorsomorphin) instead of Step 2 BMP activation will maintain the organism in $\mathcal{A}_{E1}$ (pre-bilateral disc morphology) and prevent bilateral organisation indefinitely.

Geometric basis: If the $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$ transition requires BMP deployment, then BMP inhibition after Axis 1 specification must prevent the transition. This is the negative control prediction for the two-axis attractor model.

Significance: This prediction is the experimental demonstration that $\mathcal{A}_{E1}$ is a stable attractor, not a transient state. Organisms can be maintained in $\mathcal{A}_{E1}$ indefinitely by blocking the dimensional expansion.

Failure condition: BMP inhibition does not prevent bilateral organisation after Axis 1

is specified — bilateral morphology emerges without BMP regardless.

7 Connection to the LUCA Paper

This amendment is geometrically connected to the companion publication *The Ribosome Is the LUCA* (DOI: 10.5281/zenodo.18988844).

The LUCA paper establishes that the FECA→LECA mitochondrial event is a dimensional expansion of the cellular attractor space: the energy dimension opened, and all post-LECA organisms are permanently in the expanded energy attractor space.

Theorem 4.1 of this amendment establishes that the $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$ bilateral symmetry transition is also a dimensional expansion, structurally identical to the FECA→LECA event but in morphological attractor space.

Corollary 7.1 (Dimensional Expansions Are the Structural Events of Macroevolution)**.** *The geometry of the Attractor Framework identifies two classes of evolutionary event:*

(i) ***Attractor traversal:*** *the organism moves through the attractor landscape along a known trajectory. The organism changes but remains in the same dimensional space. Accessible by developmental arrest.*

(ii) ***Dimensional expansion:*** *a new independent dimension is added to the attractor space. The organism is now in a fundamentally larger space. Not reversible by arrest alone — the new dimension must be actively closed.*

The known dimensional expansions in the history of life accessible to this framework are:

- *LUCA → LECA: energy dimension (mitochondria)*
- $\mathcal{A}_{E1} \to \mathcal{A}_{E2}$*: bilateral axis dimension (BMP orthogonal to Wnt)*

All other major transitions in the evolutionary table are attractor traversals, not dimensional expansions. They are accessible by developmental arrest. Dimensional expansion events are accessible by the two-step protocol: establish the prior attractor, then activate the new dimension.

8 Updated Falsifiability Conditions

The following conditions are added to the falsifiability table of the original pre-registration (Table 4, conditions F1–F7). They are additions, not replacements. Conditions F1–F7 remain in force.

ID	Condition
F8	Wnt activation in *Trichoplax* produces bilateral symmetry without BMP deployment. Would falsify the two-axis attractor model and require the original single-step Dickinsonia Protocol to be reinstated.
F9	Orthogonal BMP deployment after confirmed Axis 1 specification in *Trichoplax* does not produce bilateral organisation. Would falsify Prediction A1-P2 and require identification of an alternative second-axis mechanism. Would not falsify the framework or the Reproduction Theorem.
F10	BMP pathway components are absent or non-functional in *Trichoplax adhaerens* despite genome annotation showing their presence. Would require switching to a cnidarian starting organism and would not falsify the framework.
F11	$\mathcal{A}_{E1}$ (disc morphology after Wnt activation) is not a stable attractor — the organism immediately transits to $\mathcal{A}_{E2}$ upon Wnt activation, or degrades, with no stable intermediate. Would require revision of the two-stage model. Would not falsify the framework or the Reproduction Theorem.

None of conditions F8–F11 has been met as of amendment date 2026-03-12. None can have been met: no experiment has been run.

9 Summary: What Changed, What Did Not

Element	Status in original pre-registration	Status after Amendment A1
Reproduction Theorem	Unchanged	Unchanged
Inverse Accessibility Theorem	Unchanged	Unchanged
LECA Resurrection Protocol	Unchanged	Unchanged
LECA predictions (P1–P5)	Unchanged	Unchanged
Cross-Kingdom Convergence Prediction	Unchanged	Unchanged
LINE-1 independent access prediction	Unchanged	Unchanged
LUCA geometric status	Unchanged	Unchanged
Falsifiability conditions F1–F7	Unchanged	Unchanged
Dickinsonia Protocol (Section 8)	Single-step (Wnt only)	**Revised: two-step (Wnt + orthogonal BMP)**
Ediacaran attractor structure	Single target ($\mathcal{A}_{E2}$)	**Two stages: $\mathcal{A}_{E1}$ intermediate + $\mathcal{A}_{E2}$ bilateral target**
New predictions	None	**A1-P1 through A1-P4 added**
New falsifiability conditions	None	**F8–F11 added**

10 Locked Statement

This amendment is locked as of 2026-03-12.

The original pre-registration stands permanently unchanged at DOI: 10.5281/zenodo.18986790.

This amendment refines one prediction in that record: the Dickinsonia Protocol. It does not modify the framework, any theorem, any LECA prediction, or any other element of the original.

The refinement was derived geometrically. The Wnt/BMP literature was integrated into the attractor framework and produced the two-axis attractor theorem. The theorem produced the revised protocol. The revised protocol produced four new pre-specified, falsifiable predictions. All of this was completed before any experiment was run.

The geometry required bilateral symmetry to have two axes. The literature confirmed that it does. The protocol was revised to match. This is the framework working correctly.

"Bilateral symmetry is not a destination. It is a second opening. The first axis is the ground. The second axis is the expansion. The geometry always required both. Now the protocol states both explicitly."

— Eric Robert Lawson, March 12, 2026

Repository and Reproducibility

All derivation records, protocols, reasoning artifacts, pre-registration documents, and this amendment are publicly available at:

https://github.com/Eric-Robert-Lawson/attractor-oncology

Repository ID: 1172048282 | MIT License | Python | Public

Competing Interests

The author has no competing interests to declare. No institutional funding was received. No conflicts of interest exist. All data used is publicly available. All code is open source under the MIT licence.

References

[1] Holzem, M., Boutros, M., & Holstein, T.W. (2024). The origin and evolution of Wnt signalling. *Nature Reviews Genetics*, 25(7), 500–512. doi:10.1038/s41576-024-00699-w

[2] Clark, E.M., & Petersen, C.P. (2023). BMP suppresses WNT to integrate patterning of orthogonal body axes in adult planarians. *PLOS Genetics*, 19(2), e1010608. doi:10.1371/journal.pgen.1010608

[3] Mörsdorf, D., et al. (2024). Highly conserved and extremely evolvable: BMP signalling in secondary axis patterning of Cnidaria and Bilateria. *Discover Developmental Biology*, 1, 7. doi:10.1007/s00427-024-00714-4

[4] Uveira, J., et al. (2025). Planar cell polarity coordination in a cnidarian embryo provides clues to animal body axis evolution. *eLife*, e104508. doi:10.7554/eLife.104508

[5] Srivastava, M., et al. (2008). The *Trichoplax* genome and the nature of placozoans. *Nature*, 454(7207), 955–960. doi:10.1038/nature07191

[6] Bobrovskiy, I., et al. (2018). Ancient steroids establish the Ediacaran fossil *Dickinsonia* as one of the earliest animals. *Science*, 361(6408), 1246–1249. doi:10.1126/science.aat7228

[7] Lawson, E.R. (2026). Attractor Geometry of Life: Pre-Registration of Geometric Derivations, Experimental Predictions, and Falsifiability Conditions. *Zenodo*. doi:10.5281/zenodo.18986790

[8] Lawson, E.R. (2026). The Ribosome Is the LUCA: A Geometric Derivation That the Last Universal Common Ancestor Is Not Extinct, Has Never Been Extinct, and Is Currently Executing in Every Cell on Earth. *Zenodo*. doi:10.5281/zenodo.18988844

[9] Moody, E.R.R., et al. (2024). Metabolism, genome and age of the last universal common ancestor. *Nature*, 628, 545–550. doi:10.1038/s41559-024-02461-1

[10] Kalinka, A.T., et al. (2010). Gene expression divergence recapitulates the developmental hourglass model. *Nature*, 468(7325), 811–814. doi:10.1038/nature09634

[11] Irie, N., & Kuratani, S. (2011). Comparative transcriptome analysis reveals vertebrate phylotypic period during organogenesis. *Nature Communications*, 2, 248. doi:10.1038/ncomms1248